

SEQUENCE OF REAL NUMBERS

Formula & Concept Cheat Sheet

1. Basics & Definitions

Sequence: A function whose domain is the set $\mathbb{N}$ of all natural numbers and range is a subset of $\mathbb{R}$. Denoted by $\langle a_n \rangle$ or $\{a_n\}$.

Range: The set of all distinct terms of a sequence.

Bounded Sequence: A sequence $\langle a_n \rangle$ is bounded if its range set is bounded ($\exists k', k$ such that $k' \leq a_n \leq k, \forall n \in \mathbb{N}$).

- Supremum (l.u.b) and infimum (g.l.b) exist for the range set.

2. Limits & Cluster Points

Cluster Point (Limit Point): $l \in \mathbb{R}$ is a cluster point if $\forall \epsilon > 0$ and positive integer $m, \exists k > m$ such that $|a_k - l| < \epsilon$.

- The ϵ -neighborhood of l contains infinitely many terms of the sequence.

Limit of a Sequence: $l \in \mathbb{R}$ is the limit if $\forall \epsilon > 0, \exists m \in \mathbb{N}$ such that $|a_n - l| < \epsilon, \forall n \geq m$.

3. Monotonicity

- Increasing:** $a_{n+1} \geq a_n, \forall n \in \mathbb{N}$
- Decreasing:** $a_{n+1} \leq a_n, \forall n \in \mathbb{N}$
- Strictly Increasing:** $a_{n+1} > a_n, \forall n \in \mathbb{N}$
- Strictly Decreasing:** $a_{n+1} < a_n, \forall n \in \mathbb{N}$

4. Types of Sequences

Convergent: A sequence whose limit exists finitely.

- Every convergent sequence has a unique limit point.
- Every convergent sequence is bounded (converse is false).

Divergent: A monotonic and unbounded sequence. (Diverges to $+\infty$ or $-\infty$).

Oscillatory: Neither converges nor diverges.

- Finitely:* Bounded sequence not converging.
- Infinitely:* Unbounded sequence not diverging.

5. Algebra of Limits

If $\lim a_n = a$ and $\lim b_n = b$, then:

- (i) $\lim(ka_n) = ka$
- (ii) $\lim |a_n| = |a|$
- (iii) $\lim(a_n \pm b_n) = a \pm b$
- (iv) $\lim(a_n b_n) = ab$
- (v) $\lim \frac{1}{a_n} = \frac{1}{a} \quad (a \neq 0)$
- (vi) $\lim \frac{a_n}{b_n} = \frac{a}{b} \quad (b \neq 0)$

6. Important Standard Limits

- (a) If $p > 0$, then $\lim \frac{1}{n^p} = 0$
- (b) If $p > 0$, then $\lim \sqrt[p]{p} = 1$
- (c) $\lim \sqrt[p]{n} = 1$
- (d) If $p > 0$, then $\lim \frac{n^2}{(1+p)^n} = 0$
- (e) If $|x| < 1$, then $\lim x^n = 0$
- (f) $\lim \left(1 + \frac{1}{n}\right)^n = e$

7. Cauchy Sequences

Definition: $\langle a_n \rangle$ is Cauchy if for given $\epsilon > 0, \exists m \in \mathbb{N}$ such that $|a_n - a_m| < \epsilon, \forall n, m \geq m$.

Alternatively: $|a_{m+p} - a_m| < \epsilon, \forall p > 0$.

Results:

- A sequence is convergent $\iff$ it is a Cauchy sequence.
- A Cauchy sequence is always bounded.

8. Subsequences

Let $\langle a_n \rangle$ be a sequence. If $\langle n_k \rangle$ is a strictly increasing sequence of natural numbers, then $\langle a_{n_k} \rangle$ is a subsequence.

- Every sequence is a subsequence of itself.
- If $a_n \rightarrow l$, every subsequence also converges to l .
- Every sequence contains a monotonic subsequence.
- Every bounded sequence contains a convergent subsequence.

9. Limit Superior & Inferior

Let $\langle a_n \rangle$ be any sequence.

- Let $b_n = \sup\{a_n, a_{n+1}, \dots\}$. Then $\langle b_n \rangle$ is monotonically decreasing.
 $\limsup a_n = \inf\{b_1, b_2, \dots\} = \inf(b_n)$.
- Let $c_n = \inf\{a_n, a_{n+1}, \dots\}$. Then $\langle c_n \rangle$ is monotonically increasing.
 $\liminf a_n = \sup\{c_1, c_2, \dots\} = \sup(c_n)$.

Remark: For convergent sequences,
 $\limsup a_n = \liminf a_n = \lim a_n$.

10. Important Limit Theorems

Cauchy's First Theorem:

If $\lim a_n = l$, then the sequence $\langle x_n \rangle$ where

$$x_n = \frac{a_1 + a_2 + \dots + a_n}{n}$$

also converges to l . (Converse is not true).

Cauchy's Second Theorem:

If $\langle a_n \rangle$ is a sequence of positive terms, then

$$\lim(a_n)^{1/n} = \lim \frac{a_{n+1}}{a_n}$$

provided the right hand side limit exists (finite or infinite).

Cesaro's Theorem:

If $\lim a_n = a$ and $\lim b_n = b$, then

$$\lim \frac{a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1}{n} = ab$$

Sandwich Theorem:

Let a_n, b_n, c_n be sequences such that $a_n \leq b_n \leq c_n$. If $\lim a_n = \lim c_n = l$, then $\lim b_n = l$.